A Multicondition Normalized BRDF Correction Framework for Inland Water Bodies Using Airborne Hyperspectral Imagery

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***Abstract*—The bidirectional reflectance distribution function (BRDF) effects in airborne hyperspectral images can generate severe brightness gradients and radiometric inconsistency over inland water bodies, hindering precise quantitative applications. This challenge is particularly acute for lightweight airborne hyperspectral systems, where multistrip acquisition and frequent attitude fluctuations can amplify water-surface specular reflection and scattering, resulting in severe radiometric inconsistencies. We propose a unified BRDF correction framework for inland water bodies, driven by the normalization of multiple imaging conditions. By physically decoupling high-frequency sensor attitude fluctuations from periodic sensor heading systemic shifts, and constructing a kernel-based semiempirical model optimized for the complex anisotropic scattering mechanisms of inland water, this framework reduces radiometric distortions induced by dynamic attitude variations and characterizes water-body BRDF properties. By conducting an overall homogeneity visual interpretation of the mosaicked images and quantitatively calculating the reflectance of overlapping regions, we demonstrate that the correction reduces BRDF-induced radiometric inconsistency. Specifically, the mean absolute difference (MAD) of water reflectance in overlapping areas is reduced by over 50%. Evaluated against ground-measured spectra, the coefficient of determination (R^2) of the corrected reflectance reaches over 0.9. Compared with traditional BRDF correction methods, the proposed framework achieves superior performance in both visual effect and quantitative evaluation. Through comparative validation from different regions, the robustness and accuracy of the model are further demonstrated, providing precise data support for subsequent high-precision quantitative inversion and mapping. With the trend of lightweight airborne imaging systems, the proposed BRDF correction framework offers a robust radiometric baseline and has extensive application prospects for hyperspectral applications.**

***Index Terms*—Bidirectional reflectance distribution function (BRDF), hyperspectral, inland water bodies, multiple imaging conditions.**

I. Introduction

REMOTE sensing is widely used in the field of environmental monitoring because of its wide-area coverage and long-time sequence monitoring capabilities. With the development of spectral imaging technology, hyperspectral remote sensing can effectively characterize the spectral characteristics of land objects due to the high spectral resolution, thus showing potential for quantitative monitoring [1]. The bidirectional reflectance distribution function (BRDF) is an important factor used to describe the optical characteristics of surface reflectance as it relates to the physical structure of the object. Due to the anisotropic properties of objects, the observed reflectance can be influenced by the solar-sensor observation geometry, terrain, and other factors, leading to inconsistency in the imagery. For water-land scenes, water exhibits significantly different specular reflection characteristics compared with the scattering characteristics of vegetation and particles [2], [3], and land-oriented BRDF models alone are generally inadequate for water bodies. Therefore, it is necessary to separate the water bodies in water-land scenes and perform the correction separately [4], [5].

For water bodies, the causes of image inconsistency include specular reflection, water column scattering [6], and skylight scattering effects [7]. Different from traditional land BRDF correction, which mainly addresses anisotropic reflectance from relatively stable vegetation, soil, or built-up surfaces, water-specific BRDF correction must consider the combined effects of surface specular reflection, volume scattering within the water column, skylight reflection, and the low magnitude of water-leaving radiance. As a result, water BRDF effects are more sensitive to illumination-viewing geometry and platform attitude changes, especially in airborne hyperspectral images acquired by lightweight platforms. Specular reflection from water bodies often manifests as sun glint, which is caused by the solar radiation signal being reflected at the surface of water bodies observed at a specific viewing angle. In real images, sun glint appears as bright patches of light. The most commonly

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used sun glint correction methods are based on the assumption of complete absorption of light in the near-infrared (NIR) bands by water bodies [8], [9]. In addition to sun glint, water column scattering and skylight scattering effects also result in directional effects. These directional effects still exist in calm water surfaces without sun glint and cannot be mitigated using traditional sun glint correction [10]. Early studies on the anisotropy of water used physical models and Monte Carlo methods to simulate the ocean water surface and analyzed the correlations between the remote sensing reflectance of water, wavelength, viewing angle, and the inherent optical properties (IOPs) of water bodies [11]. The BRDF models based on statistical relationships have been widely applied in ocean water, but their applicability in inland water bodies is likely to be limited. The semiempirical models based on a water radiative transfer model and free parameters have also been used for water BRDF correction. However, most of the methods are focused on in situ spectra and multispectral satellite sensors [12]. Performing BRDF correction on water bodies in airborne hyperspectral images with more complex and varied imaging conditions presents a significant challenge.

The main purpose of BRDF correction of airborne hyperspectral images is to reduce the reflectance inconsistency caused by the solar-sensor observation geometry, terrain, and atmospheric factors. Therefore, the relevant studies have focused on how to establish models to normalize the reflectance to a fixed observation condition and eliminate the differences in reflectance. The commonly used models for describing the BRDF of land objects are mainly physical models, empirical models, and semiempirical models. Physical models have explicit physical meanings in their formulas, but they are typically complex and difficult to compute [13]. Empirical and semiempirical models consider the statistical features of the images and are commonly utilized for the calibration of imaging spectrometer data [14]. Recent airborne and uncrewed aerial vehicle (UAV)-based hyperspectral studies have further shown that strip-to-strip radiometric differences are closely related to flightline-level observation geometry, illumination variation, and overlap consistency [15], [16]. However, most of these studies mainly focus on land-dominated scenes or general airborne mosaicking, whereas opposite-heading-induced baseline shifts over inland water bodies remain insufficiently considered. With the miniaturization and lightweight design of imaging spectrometers, in addition to fixed-wing crewed platforms, imaging spectrometers can now be carried on helicopter platforms as well as uncrewed aircraft platforms. During the imaging process, the sensor's attitude is often dynamically changing due to factors such as airflow disturbance, aircraft acceleration and deceleration, and flight attitude changes. Compared with conventional high-altitude platforms, lightweight airborne platforms generally have lower mass, lower flight altitude, and weaker attitude stability, making them more sensitive to local airflow disturbances and platform maneuvering. Especially for helicopters and UAVs, which need to change their pitch and roll angle to control acceleration/deceleration and vehicle balance, attitude fluctuation occurs frequently during the flight process. Therefore, for push-broom imaging spectrometers, large attitude differences may occur between adjacent scan lines. These scan-line-level attitude variations tend to exacerbate BRDF effects, resulting in severe radiometric inconsistency and posing significant challenges to correction.

However, most existing correction methods—often designed for relatively stable, high-altitude platforms—exhibit significant limitations when confronted with the frequent sensor attitude changes and rapidly varying imaging geometries typical of lightweight platforms. Traditional approaches are insufficient to correct the complex radiometric inconsistencies caused by abrupt attitude fluctuations. This challenge is particularly acute for water bodies, where strong specular reflection and hot-spot effects are highly sensitive to viewing angle changes. Furthermore, the opposite flight directions between adjacent strips induce drastic shifts in observation geometry, further intensifying interstrip BRDF differences. These critical factors are typically not fully incorporated into existing correction frameworks, making it difficult to achieve unified processing across different flight strips. Therefore, effective strategies that combine intrastrip attitude correction with interstrip geometric normalization for comprehensive BRDF mitigation remain to be explored.

In summary, BRDF effects are commonly present in airborne hyperspectral imagery, especially for inland water bodies surveyed by lightweight platforms where unstable imaging conditions degrade data quality. It is therefore necessary to normalize the different imaging conditions to mitigate the radiometric inconsistency caused by the BRDF effects. This article focuses on BRDF effects over inland water bodies arising from sun glint, solar azimuth angle, flight azimuth angle, and high-frequency sensor attitude changes. To address these coupled effects, we develop a multicondition normalized correction framework that accounts for sun glint, illumination geometry, flight direction, and attitude-induced observation-angle variations using semiempirical kernel functions. The framework achieves scanline-by-scanline correction within a single strip and supports image correction and seamless mosaicking across multiple strips. Demonstrating high robustness and generalizability, the proposed model is applicable to images acquired under diverse imaging conditions across various platforms.

The main contribution of this article is threefold.

1) A per-pixel observation-angle solution and normalization correction are implemented to address the real-time change of sensor flight attitude during hyperspectral image acquisition, thereby effectively mitigating brightness gradients and strip artifacts induced by frequent attitude jitter and complex viewing geometries on lightweight airborne platforms.
2) A BRDF correction framework for water-dominated hyperspectral image scenes with normalized multiple imaging conditions is proposed, which jointly accounts for sun glint, solar azimuth, flight direction, and sensor attitude, addressing the severe directional anisotropy and radiometric inconsistency.
3) The correction framework is unified and robust for radiometric consistency restoration. It can be applied

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across different sensors and platforms, generally outperforming traditional empirical and semiempirical methods in reducing brightness gradients and interstrip inconsistencies and providing high-quality reflectance data.

The remainder of this article is organized as follows. Section II reviews related work. In Section III, we present the framework of the correction model. Section IV demonstrates the performance of the proposed model, with comparative results and discussion based on different datasets. Finally, Section V concludes this article.

II. Related Works

In this section, we review three BRDF correction methods, including physical models, empirical models, and semiempirical models.

A. Physical BRDF Correction Models

Physical BRDF models are commonly used to determine the specific distribution of reflected radiance from objects, enabling the reconstruction of realistic images of objects [17]. Commonly used methods include geometric optical models [13], [18], layered models [19], microfacet models [20], and radiative transfer models [21]. However, physical models require accurate estimation of various parameters, such as surface roughness, scattering properties, and illumination conditions. These parameters are difficult to obtain accurately and require a significant amount of measured data, which can lead to inaccuracies in the corrected reflectance. Consequently, due to the limited availability of measured data, physical models often fail to achieve satisfactory correction results in large-scale imagery.

B. Empirical BRDF Correction Models

Empirical models utilize free parameters and typically consider the statistical features of the images. These models often use fitting techniques related to the observation values for calibration, such as the least-squares fitting model and the polynomial fitting model [20]. The models can be adjusted and optimized for different applications to accommodate the anisotropy of various objects. In addition, some studies have implemented multiple-regression-based correction schemes that perform BRDF correction separately for preclassified wetland vegetation and water bodies, thereby further adapting the model to land–water heterogeneity [4]. However, empirical models usually simplify the influence of various factors on radiation information, often overlooking the directional reflective properties of different land surfaces. Therefore, under varying imaging conditions, the model often lacks generalization capability.

C. Semiempirical BRDF Correction Models

The semiempirical models combine a physical model with empirical parameters to achieve high-precision calibration. The linear kernel-driven model is one of the most representative semiempirical models and has been widely used for the correction of airborne hyperspectral images. A Ross–Li kernel-based BRDF correction model using image consistency criteria to correct wide-field airborne hyperspectral images was developed [22]. The FlexBRDF correction model accounts for the effects of terrain, land cover, and solar elevation angle. This model was constructed using a combination of Ross and Li kernels and was successfully applied to correct Airborne Visible Infrared Imaging Spectrometer (AVIRIS) image data [15]. The Ross-Thick-Maignan and Li-Transit-Reciprocal kernels, combined with topographic correction methods, were utilized to correct BRDF effects in airborne hyperspectral imagery of rugged topography [23]. Land–water correction in coastal areas was achieved using FlexBRDF and sun glint correction, which effectively mitigated BRDF effects over land surfaces and sun glint effects over water [5]. However, the nonglint directional effects in water bodies were not addressed. Recently, some studies have incorporated illumination variations into correction methods, constructing correction factors to eliminate the effects of illumination differences [16], [24].

BRDF can provide robust, large-area surface constraints. However, how to reasonably design correction models that accurately and comprehensively describe the contributions of differentiated surface absorption and scattering characteristics to the BRDF remains a challenge that semiempirical models need to address.

III. Methodology

A. Image Preprocessing

Before performing BRDF correction, a series of preprocessing steps were required to obtain the surface reflectance of the imagery. The first step required nonuniformity correction and absolute radiometric calibration using laboratory integrating sphere data to obtain accurate radiance data [25]. After obtaining the radiance data, the MODTRAN radiative transfer model using the atmospheric parameters obtained using sunphotometer and meteorological station data during the imaging process was utilized to conduct atmospheric correction to obtain the reflectance (ρ) data. To isolate water bodies for correction, the normalized difference water index (NDWI) and Otsu thresholding were employed for water extraction. For the water bodies, the field-measured water reflectance was the remote sensing reflectance (R_{rs}). Its rigorous physical definition is the ratio of water-leaving radiance (L_w) to the downwelling irradiance (E_d) just above the surface, and it has units of sr^{-1}. The image after atmospheric correction is surface reflectance, so it was necessary to convert the image reflectance into remote sensing reflectance in the subsequent quantitative evaluation process. Ignoring the skylight effect, the reflectance of water ρ_{water} was converted into R_{rs} by dividing by π [26].

As sun glint can severely skew the subsequent angular modeling, a widely used NIR-based linear correlation method was applied to preliminarily eliminate high-intensity specular glint [9]. The correction formula was constructed as follows:

$$\rho_i' = \rho_i - b_i \left[\rho\,(NIR) - \rho_{\min}\,(NIR)\right] \tag{1}$$

where ρ_i is the original reflectance of the pixel to be corrected in band i, $\rho(NIR)$ is the pixel's reflectance in the NIR band, and

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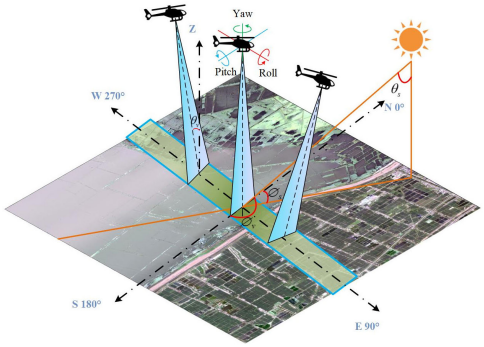


Fig. 1. Observation geometry at the imaging moment.

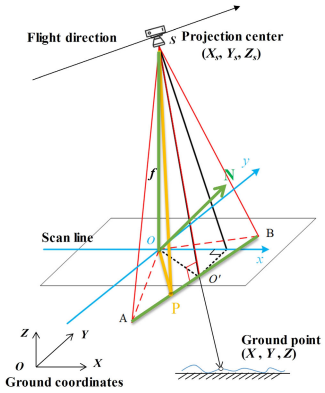


Fig. 2. Observation geometry of each pixel.

$\rho_{\min}(NIR)$ is the minimum reflectance in the NIR band, which represents the ambient NIR reflectance of a pixel without sun glint. To mitigate the impact of sediment or particulate matter on the NIR radiation signals, we used the minimum values between 970 and 980 nm for the modeling [27]. The slope b_i is the linear correlation calculated by the least-squares method between $\rho(NIR)$ and ρ_i in each visible band.

B. Observation Geometry

In the flight process, considering the variation of the sensor attitude, the observation geometry of different angles and the schematic of the observation plane at the imaging moment are shown in Fig. 1.

The change of flight attitude will lead to a change of observation angle at different imaging moments. From Fig. 1, it can be seen that the heading angle mainly affects the overall view azimuth angle of the image, and the pitch and roll angles mainly affect the view zenith angle of the image. Considering the push-broom imaging method of the imaging spectrometer, the angle calculation of each charge-coupled device (CCD) is required, and all three attitude angles contribute to the pixel view azimuth angle (φ_v) and view zenith angle (θ_v) [15].

A schematic of θ_v and φ_v is given in Fig. 2, where OS is the zenith direction, and SP is the line between pixel P on the focal plane AB and the projection center S. Considering the difference between the position vector of the CCD and the projection center, the angle $\angle OSP$ between the zenith and the line SP is the view zenith angle of pixel P, denoted as θ_v. The angle $\angle PON$ between SP and the north direction ON is the view azimuth angle of pixel P, denoted as φ_v. Considering the impact of terrain, the observation geometry needs to be refined by incorporating the slope and aspect from a digital elevation model (DEM) [28]. However, as the study area primarily consisted of a lake with flat terrain, the local solar and viewing observing angles were considered equivalent to those for flat ground.

C. Correction Method With Normalized Multiple Imaging Conditions

The proposed correction method is based on semiempirical kernel-based BRDF correction, which considers the normalization correction of multiple imaging conditions. The final normalized reflectance ρ_N for a given observation geometry (θ_v, θ_s, $\Delta\varphi$, s) at wavelength λ can be expressed as

$$\begin{aligned}\rho_N(\theta_v, \theta_s, \Delta\varphi, s, \lambda) &= \rho'(\theta_v, \theta_s, \Delta\varphi, s, \lambda) \cdot \Psi_{att}\left(\theta_{roll}, \theta_{pitch}, \lambda\right) \\ &\quad \cdot \Psi_{BRDF}(\theta_v, \theta_s, \Delta\varphi, s, \lambda)\end{aligned} \tag{2}$$

where ρ' is the water reflectance after preprocessing, $\Delta\varphi$ is the relative azimuth angle, s refers to the number of strips, and $\Psi_{att}(\theta_{roll}, \theta_{pitch}, \lambda)$ represents the sensor attitude normalization coefficient driven by the dynamic sensor roll angle θ_{roll} and pitch angle θ_{pitch}, which eliminates radiometric distortion caused by instantaneous attitude variations. $\Psi_{BRDF}(\theta_s, \theta_v, \Delta\phi, s, \lambda)$ is the semiempirical kernel correction coefficient for strip s, which normalizes the relative viewing geometry across the field of view (FOV) to a stable angle and compensates for the absolute systemic baseline shift induced by opposite flight headings. The detailed solution processes for these two correction terms are described below.

1) Sensor Attitude Normalization Correction: During the flight process, assuming that the consistency of the water bodies does not change, in the case of a constant flight attitude, the radiation signal obtained between the different scan lines should be consistent. However, the change in sensor attitude results in a significant brightness change between adjacent scan lines. The reason for this brightness change is mainly due to the weak radiation signal of the water bodies and the change in attitude intensifying the scattering effect and the adjacency effect. To address this issue, we propose a sensor attitude normalization correction method. The overall correction workflow is shown in Fig. 3.

Considering that the roll angle θ_{roll} and pitch angle θ_{pitch} are related to the sensor zenith angle θ_z to reduce the variables in the fitting function, the reflectance can be approximated as a function of θ_z. The sensor zenith angle θ_z is calculated as follows:

$$\theta_z = \arccos\left(\cos\theta_{roll}\cos\theta_{pitch}\right). \tag{3}$$

The effect of sensor attitude change on the reflectance is modeled by the reflectance corresponding to sensor zenith angles. As shown in (3), the variable θ_z is correlated with the sensor attitude of each scan line. To mitigate potential errors from individual pixels, the mean reflectance of the scan lines $[\bar{\rho}(\theta_z)]$ is used for modeling. Here, θ_z serves as the independent

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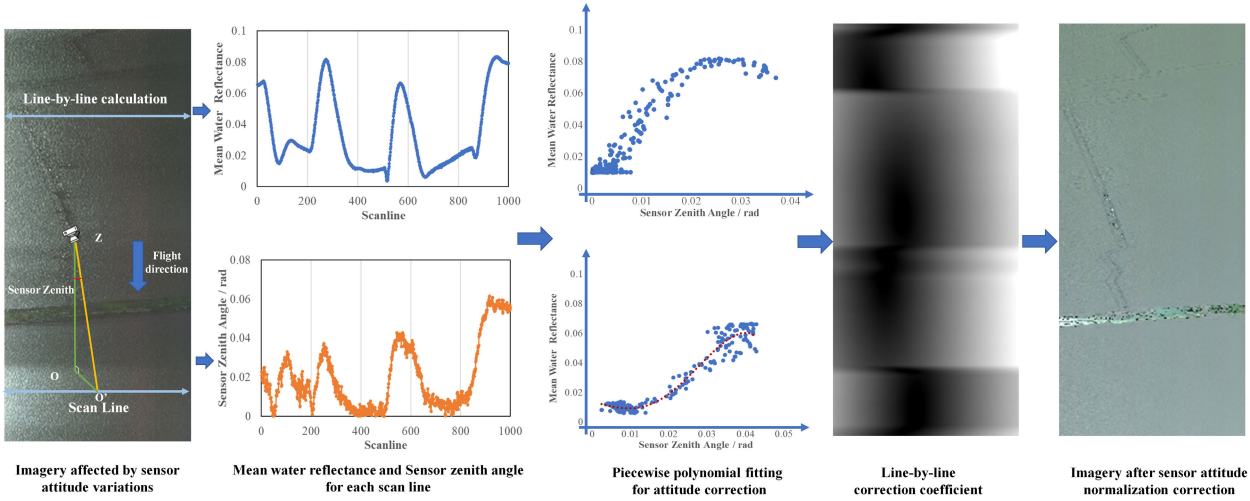


Fig. 3. Workflow of the sensor attitude normalization.

variable, while $[\bar{\rho}(\theta_z)]$ serves as the dependent variable. A quartic polynomial fitting model, selected as a low-order nonlinear model to capture the asymmetric reflectance–zenith-angle relationship while limiting the overfitting risk of higher degree polynomials, is constructed for each spectral band as follows:

$$\hat{\rho}(\theta_z) = a_0 + a_1\theta_z + a_2\theta_2^2 + a_3\theta_3^3 + a_4\theta_4^4 \tag{4}$$

where $\hat{\rho}(\theta_z)$ is the fit mean reflectance for a given sensor zenith, and a_0–a_4 are the least-squares fitting coefficients. Considering the periodic change in the sensor attitude and the inherent difference of the reflectance between different areas of water bodies, the segmented fitting process is carried out according to the attitude fluctuation. Specifically, local turning points of the sensor zenith angle profile are used as the segmentation boundaries, so that each segment corresponds to a continuous attitude fluctuation interval. The segmentation interval is set according to the sensor zenith angle, and the line-by-line correction coefficient $S(\theta_z)$ is obtained as follows:

$$S(\theta_z) = \hat{\rho}(\theta_z)/\hat{\rho}\left(\theta_z^0\right) \tag{5}$$

where $\hat{\rho}(\theta_z^0)$ is the fit reflectance evaluated at the reference sensor zenith angle of 0°. Since the radiometric information of the water bodies is affected by multiple factors, and considering the existence of continuity of the features between the scanned lines, the Savitzky–Golay filter is performed on the line-by-line correction coefficient to reduce the residual noise caused by the segmented fitting error

$$\hat{S}_j = \frac{\sum_{k=-r}^{r} C_k S_{j+k}}{2r+1} \tag{6}$$

where S and $\hat{S}$ are the original and filtered coefficient profiles, j is the running index, r is the smoothing window radius, and C_k is the weight for the kth mean value in the smoothing window. Finally, the attitude normalization coefficient Ψ_{att} in (2) can be expressed as

$$\Psi_{att} = \frac{1}{\hat{S}_j}. \tag{7}$$

After completing the attitude normalization correction, the observation geometry used for the subsequent BRDF coefficient calculation only considers sensor FOV and the yaw angle (θ_{yaw}). Finally, θ_v and φ_v can be expressed as follows:

$$\theta_v(x) = \theta_{FOV}\frac{\left(x - \frac{Samples}{2}\right)}{Samples} \tag{8}$$

$$\varphi_v\left(x, \theta_{yaw}\right) = \begin{cases} \theta_{yaw} - \dfrac{\pi}{2}, & x < \dfrac{Samples}{2} \\ \theta_{yaw} + \dfrac{\pi}{2}, & x \geq \dfrac{Samples}{2} \end{cases} \tag{9}$$

where x is the position of the pixel in the scan line, *Samples* is the total number of samples of the sensor, and θ_{FOV} is the sensor FOV.

2) Kernel-Based Semiempirical Inland Water Correction Model: After decoupling radiometric distortion caused by attitude variations, the directional reflection properties of inland water bodies are characterized using a semiempirical approach. Inland water bodies not only exhibit volumetric scattering but are also accompanied by significant anisotropic specular reflection on the surface. In this study, we established a kernel-based semiempirical inland water correction model to accurately characterize the forward scattering characteristics of water bodies by integrating the Cox–Munk kernel with the traditional Ross-Thick volumetric kernel and Li-Sparse geometric kernel. The integrated Cox–Munk kernel, physically derived from Fresnel's law of reflection and the surface slope distribution model, can characterize the forward scattering properties of water bodies [29], [30]. The Cox–Munk BRDF model can be expressed as follows [31]:

$$\mathrm{BRDF}(\theta_v, \theta_s, \Delta\varphi, n) = \frac{r_f p(\theta_n) d(\theta_v, \theta_s, \sigma)}{4\cos\theta_v \cos\theta_s \cos^4\theta_n} \tag{10}$$

where r_f is the Fresnel reflection coefficient, driven by the water refractive index, p is the wave slope distribution function, θ_n is the polar angle of the normal vector of the wave facet, and d is the shadow factor that considers the occlusion of each facet. The BRDF model was calculated for each spectral band as follows:

$$\rho_i(\theta_v, \theta_s, \Delta\varphi) = f_{iso,i} + f_{vol,i}K_{vol}(\theta_v, \theta_s, \Delta\varphi) + f_{\mathrm{geo},i}K_{\mathrm{geo}}(\theta_v, \theta_s, \Delta\varphi)$$

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$$+ f_{\text{spec},i} K_{\text{spec}}(\theta_v, \theta_s, \Delta\varphi) \tag{11}$$

where ρ_i is the reflectance in band i, $\Delta\varphi$ is the relative azimuth angle, and K_{vol}, K_{geo}, and K_{spec} are the volumetric, geometric, and Cox–Munk kernels, respectively. f is the model coefficient, which is derived by the least-squares method. For different ground objects, the model coefficient $f_{kernel,i}$ in band i is calculated as follows:

$$F = \left(X^{\mathrm{T}}X\right)^{-1} X^{\mathrm{T}}Y \tag{12}$$

where

$$X = \begin{bmatrix} 1 & K_{vol1} & K_{geo1} & K_{spec1} \\ 1 & K_{vol2} & K_{geo2} & K_{spec2} \\ \cdots & \cdots & \cdots & \cdots \\ 1 & K_{voln} & K_{geon} & K_{specn} \end{bmatrix}, \quad Y = \begin{bmatrix} \rho_{1,i} \\ \rho_{2,i} \\ \cdots \\ \rho_{n,i} \end{bmatrix}, \quad F = \begin{bmatrix} f_{iso,i} \\ f_{vol,i} \\ f_{\text{geo},i} \\ f_{\text{spec},i} \end{bmatrix}.$$

The detailed calculation of the kernel functions is as follows:

$$K_{\text{geo}} = O(\theta_s, \theta_v, \Delta\varphi) - \sec\theta'_s - \sec\theta'_v + \frac{1}{2}\left(1 + \cos\xi''\right)\Big) \tag{13}$$

$$K_{\text{vol}} = \frac{\left(\frac{\pi}{2} - \xi\right)\cos\xi + \sin\xi}{\cos\theta_s + \cos\theta_v} - \frac{\pi}{4} \tag{14}$$

$$K_{spec} = \begin{cases} \dfrac{1}{\cos\theta_s}\left(1 - \dfrac{\tan^2\theta_n}{\sigma^2}\right) - 1, & \frac{\tan^2\theta_n}{\sigma^2} \le 1 \\ -1, & \frac{\tan^2\theta_n}{\sigma^2} > 1 \end{cases} \tag{15}$$

$$O = \frac{1}{\pi}(t - \sin t\cos t)\left(\sec\theta'_s + \sec\theta'_v\right) \tag{16}$$

$$\cos t = \frac{h}{b}\frac{\sqrt{D^2 + \left(\tan\theta'_s \tan\theta'_v \sin\Delta\varphi\right)^2}}{\sec\theta'_s + \sec\theta'_v} \tag{17}$$

$$D = \sqrt{\tan^2\theta'_s + \tan^2\theta'_v - 2\tan\theta'_s \tan\theta'_v \cos\Delta\varphi} \tag{18}$$

$$\theta' = \tan^{-1}\left(\frac{b}{r}\tan\theta\right) \tag{19}$$

$$\cos\xi' = \cos\theta'_s\cos\theta'_v + \sin\theta'_s\sin\theta'_v\cos\Delta\varphi \tag{20}$$

$$\cos\xi = \cos\theta_s\cos\theta_v + \sin\theta_s\sin\theta_v\cos\Delta\varphi \tag{21}$$

$$\cos^2\theta_n = \frac{(\cos\theta_s + \cos\theta_v)^2}{2 + 2\cos\Delta\varphi} \tag{22}$$

$$\sigma^2 = 0.003 + 0.00512\omega \tag{23}$$

where h/b and b/r are the structural parameters describing the crown relative height and shape. The constant values are $h/b = 2$ and $b/r = 1$ derived in the MODIS BRDF/albedo algorithm. ω is the measured wind speed (m/s).

Traditional workflows typically utilize the anisotropy factor (ANIF) to normalize the observation geometry to a stable reference zenith plane. In actual flight missions, to maximize flight efficiency, the flight direction between adjacent air routes may undergo a 180° reversal (for example, the first route flies from east to west, while the second route flies from west to east). This can lead to extreme nonlinear changes in $\Delta\varphi$. Opposite flight directions induce substantial changes in the scattering angular distribution of water-leaving and skylight radiance. In addition, specular reflection and hot-spot effects over water bodies further enlarge interstrip reflectance differences. These effects introduce additional errors when solving BRDF model parameters with multiple strip inputs, making it impossible to maintain a global absolute radiometric baseline solely through local kernel functions. We therefore embed macroscopic heading scattering constraints into the reference frame of anisotropy calculation, defining a systematic-constrained ANIF (SC-ANIF). Specifically, the flight strip $s-1$ with the minimal influence of solar radiation and attitude distortion is selected as the reference. Based on the principle of radiation consistency in geometric overlapping regions, a systematic constrained coefficient C_s is formulated as a robust physical boundary condition against systemic and heading-induced scattering drift. For any pixel in strip s, the SC-ANIF can be expressed as follows:

$$\begin{aligned} \text{SC-ANIF} &= \frac{BRF_C\left(\theta'_v, \theta'_s, \Delta\varphi\right)}{C_s \cdot BRF_N\left(\theta^0_v, \theta^0_s, \Delta\varphi^0\right)} \\ &= \frac{\bar{\rho}_{s,overlap} \cdot \rho\left(\theta'_v, \theta'_s, \Delta\varphi\right)}{\bar{\rho}_{s-1,overlap} \cdot \rho_0\left(\theta^0_v, \theta^0_s, \Delta\varphi^0\right)} \end{aligned} \tag{24}$$

where ρ is the fitting reflectance under an arbitrary solar-sensor observation geometry, and ρ_0 is the fitting reflectance at a fixed observation geometry. $\bar{\rho}_{s,overlap}$ is the average reflectance of the overlapping area. The fixed reference observation zenith angle is 0°, and the fixed solar zenith angle θ^0_s and relative azimuth angle $\Delta\varphi^0$ are determined based on the sun's position at the imaging time. Finally, the semiempirical kernel correction coefficient Ψ_{BRDF} in (2) can be expressed as

$$\Psi_{BRDF} = \frac{1}{\text{SC-ANIF}}. \tag{25}$$

IV. Experimental Results and Analysis

A. Datasets

In this study, the hyperspectral image datasets were collected over two different lakes—Gaoyou Lake and Dianshan Lake—in the middle and lower reaches of the Yangtze River in China. For Gaoyou dataset, the image data were obtained using an HMS400A [32] sensor mounted on a helicopter platform. The data collection took place on August 24, 2020, at Gaoyou Lake, with a time span from 10:00 to 13:30, and a total of 18 strips of images were obtained. The imagery exhibits significant sun glint and nonuniformity due to the midday acquisition time and abrupt changes in helicopter attitude during flight.

For Dianshan dataset, the image data were obtained using an AMMIS sensor mounted on a fixed-wing aircraft. The imagery was collected on June 15, 2022 and June 16, 2022, with a time span from 08:30 to 10:30, and a total of 15 strips of images were obtained. Taking into account the solar zenith angle at the flight time, sun glint was largely avoided. Meanwhile, fixed-wing aircraft also have better stability and can avoid sudden changes in flight attitude.

The imaging system utilized in both regions was made up of an imaging spectrometer, a stable platform, and an inertial navigation unit. The imaging spectrometer parameters are listed in Table I. In order to ensure that the imagery and spectral information acquired by the hyperspectral sensors met the experimental requirements, the sensors were tested and calibrated before use, including wavelength calibration and a

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TABLE I
SPECTROMETER PERFORMANCE PARAMETERS

Parameter	HMS400A	AMMIS
Spectral range/nm	400–1000	400–1000
Spectral resolution/nm	1.5–4.0	5.0
Bands	256	256
Samples	2048	3000
IFOV/mrad	1.2	0.25
FOV/°	34	40

sensitivity test. Water spectra and typical ground object spectra were collected simultaneously using an ASD FieldSpec3 spectrometer during the flight for validation within a 5–10-min window of the corresponding image acquisition.

B. Methodological Comparison

1) Quartic Empirical Correction Method: To verify the performance of the proposed correction method, we compared three correction methods. The quartic empirical correction method is used to establish the quartic function relationship between the mean column reflectance and the corresponding view zenith angle and then normalize the reflectance from different viewing angles to a common reference geometry (the minimum view zenith angle) [25].

2) Traditional Kernel-Based Correction Method: The main difference between the proposed method and the traditional kernel-based correction method lies in the sensor attitude normalization and strip matching. To verify the effectiveness of the proposed method, the traditional kernel-based correction method was applied, consistent with the proposed method, but without normalization correction for multiple imaging conditions.

3) Semiempirical Water BRDF Correction Method: We also compared the semiempirical water BRDF correction method proposed for water bodies [33]. This method considers the different effects of molecular scattering and calculates R_{rs} using IOPs and independent model variables.

C. Assessment

1) Qualitative Evaluation: The qualitative evaluation was performed by visually judging whether there were obvious brightness or color differences in the images and also by observing the consistency of the overall image color after image mosaicking and whether there were obvious seam lines.

2) Relative Quantitative Evaluation: The relative quantitative evaluation was performed for the overlapping areas between the strips and was quantified by calculating the relative reflectance differences. The relative evaluation indicator was the mean absolute difference (MAD) in the overlapping areas [4].

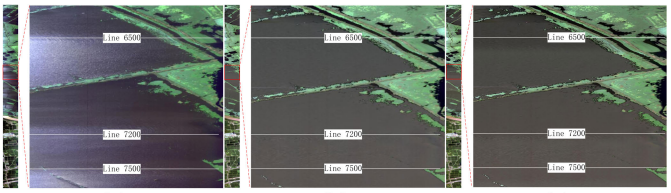
3) Absolute Quantitative Evaluation: The accuracy of the image reflectance directly affects the accuracy of the subsequent applications. The absolute evaluation indicators of the mean square error (mse), determination coefficient (R^2), and spectral angle (SA) were utilized to evaluate the accuracy between the image reflectance after correction and the ground-truth reflectance

$$\text{mse} = \frac{\sum_{i=1}^{n} (\hat{y}_i - y_i)^2}{n} \tag{26}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (\hat{y}_i - y_i)^2}{\sum_{i=1}^{n} (\bar{y}_i - y_i)^2} \tag{27}$$

$$\text{SA} = \cos^{-1}\left(\frac{\vec{\mathbf{t}} \cdot \vec{\mathbf{r}}}{\|\vec{\mathbf{t}}\| \times \|\vec{\mathbf{r}}\|}\right) = \cos^{-1}\left(\frac{\sum_{i=1}^{b} t_i r_i}{\sqrt{\sum_{i=1}^{b} t_i^2} \times \sqrt{\sum_{i=1}^{b} r_i^2}}\right) \tag{28}$$

where y_i is the measured reflectance value, $\hat{y}_i$ is the image reflectance value, n is the number of samples, b is the number of bands, $\vec{\mathbf{t}}$ is the image spectrum, and $\vec{\mathbf{r}}$ is the ground-truth spectrum. The lower the mse and SA values and the higher the R^2 value, the higher the accuracy. An R^2 value greater than 0.9 and an SA value less than 0.3 rad (17.2°) can be considered a high degree of similarity.



(a) (b) (c)

Fig. 4. Single-strip images before and after correction. RGB color composition with 639.51, 549.99, and 460.48 nm. (a) Before correction. (b) After correction. (c) Correction without sensor attitude normalization.

D. Results

1) Single-Strip Correction Results: In order to verify the effectiveness of the proposed method in mitigating the radiometric inconsistency between adjacent scan lines caused by sensor attitude changes, we analyzed the results before and after the correction of a single strip with abrupt attitude changes in Gaoyou Lake dataset. We chose a strip that had not been geometrically corrected, in order to determine the corresponding scan line and its attitude information when the image brightness changes. The original uncorrected image is shown in Fig. 4(a). It can be seen that there are obvious changes in brightness near lines 6500, 7200, and 7500. By analyzing the flight attitude of the scan lines, as shown in Fig. 5, it is found that there are obvious pitch and roll angle changes at lines 6500, 7200, and 7500. Therefore, the image before correction was affected by the change of flight attitude, resulting in obvious brightness differences between adjacent scan lines. This difference was not caused by changes in the properties of the water body itself. The corrected image mitigates the obvious brightness differences between adjacent lines, as shown in Fig. 4(b). In addition, to further verify the effectiveness of sensor attitude normalization, we conducted a

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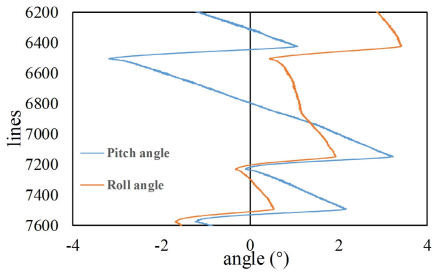


Fig. 5. Flight attitude of the scan lines.

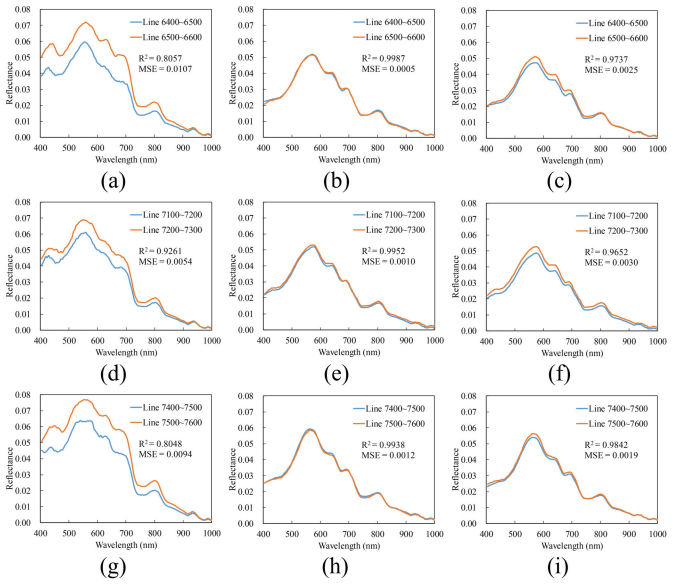


Fig. 6. Reflectance comparison of adjacent scan lines before and after correction. (a), (d), and (g) Average reflectance curves of adjacent scan lines before correction. (b), (e), and (h) Reflectance curves of adjacent scan lines after correction. (c), (f), and (i) Reflectance curves of adjacent scan lines after correction without sensor attitude normalization.

comparative experiment using correction without sensor attitude normalization, as shown in Fig. 4(c). It can be observed that although the subsequent processing steps (without sensor attitude normalization) can eliminate most of the sun glint and mitigate the overall nonuniformity to some extent, obvious radiometric discrepancies remain between adjacent scanlines where attitude changes occurred.

In order to quantitatively evaluate the changes in reflectance, we selected 100 scan lines of water bodies before and after the occurrence of obvious pitch angle changes and calculated their average reflectance values. The contrast curves of the reflectance between adjacent scan lines are shown in Fig. 6. By comparing the reflectance curves and calculating the R^2 and mse values of adjacent scan lines, it can be seen that there was a significant difference in reflectance before correction, which can reach 20%–30%. After image correction without sensor attitude normalization, the differences in reflectance were reduced, but an error of about 5% still remained. After image correction by the sensor attitude normalization method proposed in this article, the radiation nonuniformity between adjacent scan lines caused by the sensor attitude is significantly improved, and the difference in reflectance is less than 3%.

2) Image Mosaicking Results: In order to visualize the image differences before and after correction, an image mosaic of the original image strips, the quartic empirical corrected image, the traditional kernel-based corrected image, the image corrected by the semiempirical water BRDF correction method, and the proposed BRDF-corrected image was compared. The mosaicking results for Gaoyou dataset are shown in Fig. 7.

In Gaoyou dataset, due to the presence of a serious sun glint phenomenon in the original image strips, the brightness difference between strips is significant, and the image consistency of the water body is severely damaged, as shown in Fig. 7(a). The three comparison correction methods are able to mitigate the inconsistencies between strips. In Fig. 7(b)–(d), it can be seen that, compared with the original image, the brightness gradient between strips has been significantly improved, but the correction is less effective in areas that are seriously affected by sun glint. In addition, the three correction results do not significantly improve the brightness differences between adjacent scan lines caused by sensor attitude on the same flight strip. The proposed method shown in Fig. 7(e) can effectively mitigate the phenomenon of sun glint and brightness gradient in the same flight strip. The inconsistency between adjacent strips is significantly mitigated, achieving the best mosaicked image consistency. We extracted vertical reflectance profiles for the red, green, and blue bands over a homogeneous water region, as shown in Fig. 8. The profiles obtained using the proposed method exhibit the smallest fluctuations across stripe boundaries, demonstrating that the interstrip reflectance inconsistencies have been effectively corrected.

We also applied the proposed correction model to Dianshan dataset. The mosaicking results are shown in Fig. 9. Although there is no severe sun glint phenomenon in Dianshan Lake image strips, there are still significant brightness differences due to the water column scattering and skylight scattering effects of the water bodies, as shown in Fig. 9(a). In addition, there is a strong residual strip boundary toward the bottom of the image above the canal. All three comparison methods can effectively mitigate the directional reflection of water, and the consistency of the images is significantly improved. However, there are still significant brightness differences in the bottom of the mosaicked image. The proposed method takes into account the difference in flight direction of strips and performs strip matching correction, resulting in the best mosaicked image consistency, as shown in Fig. 9(e). The residual stripe boundaries at the bottom of the image have also been significantly eliminated. We also extracted vertical reflectance profiles for the red, green, and blue bands over a homogeneous water region in Dianshan Lake, as shown in Fig. 10. The results indicate that the proposed method achieves the best consistency among all compared approaches.

E. Accuracy Assessment Results and Discussion

1) Gaoyou Dataset: Fig. 11(a) shows the MAD curves of each band before and after correction in Gaoyou dataset. The sun glint in the uncorrected images results in higher MAD values in the overlapping areas, with a maximum MAD value of 0.198. The method proposed in this article shows a significant correction effect on the water bodies, with a maximum MAD value of 0.0057, which is more than 95% lower than that of the uncorrected image strips. In addition,

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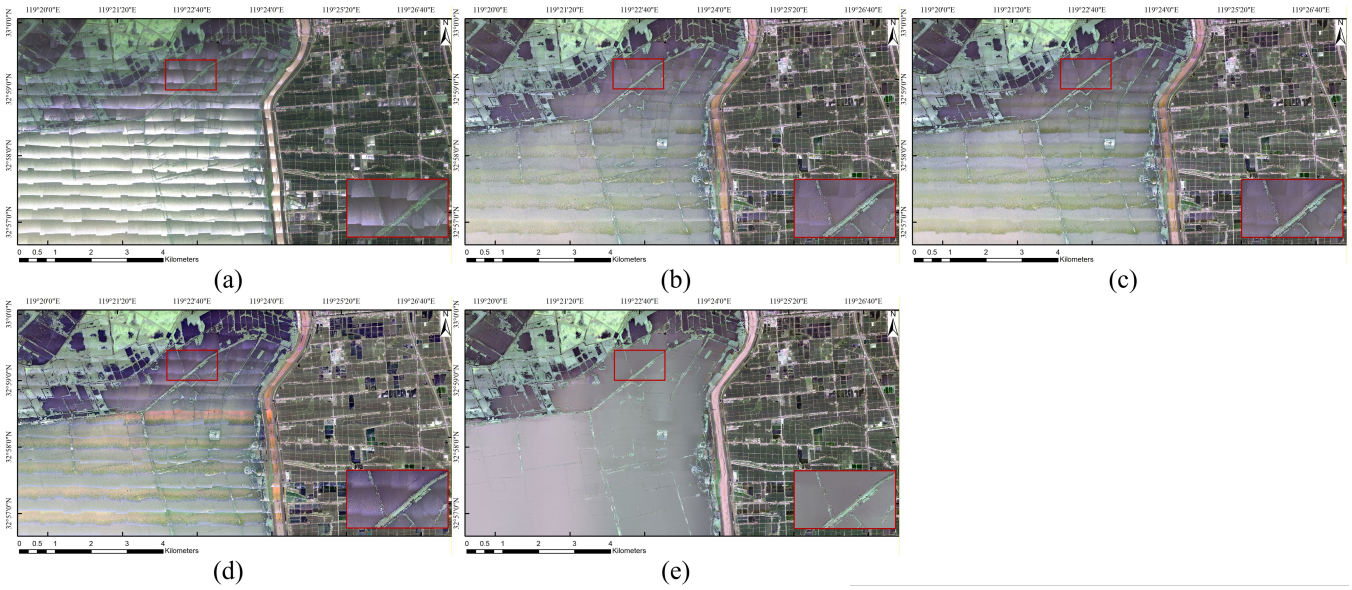


Fig. 7. Image mosaicking results for the 18 image strips of Gaoyou Lake. RGB color composition with 639.51, 549.99, and 460.48 nm. (a) Mosaic result for the original image strips. (b) Mosaic result after quartic empirical correction. (c) Mosaic result after traditional kernel-based correction. (d) Mosaic result after semiempirical water BRDF correction. (e) Mosaic result using the proposed method.

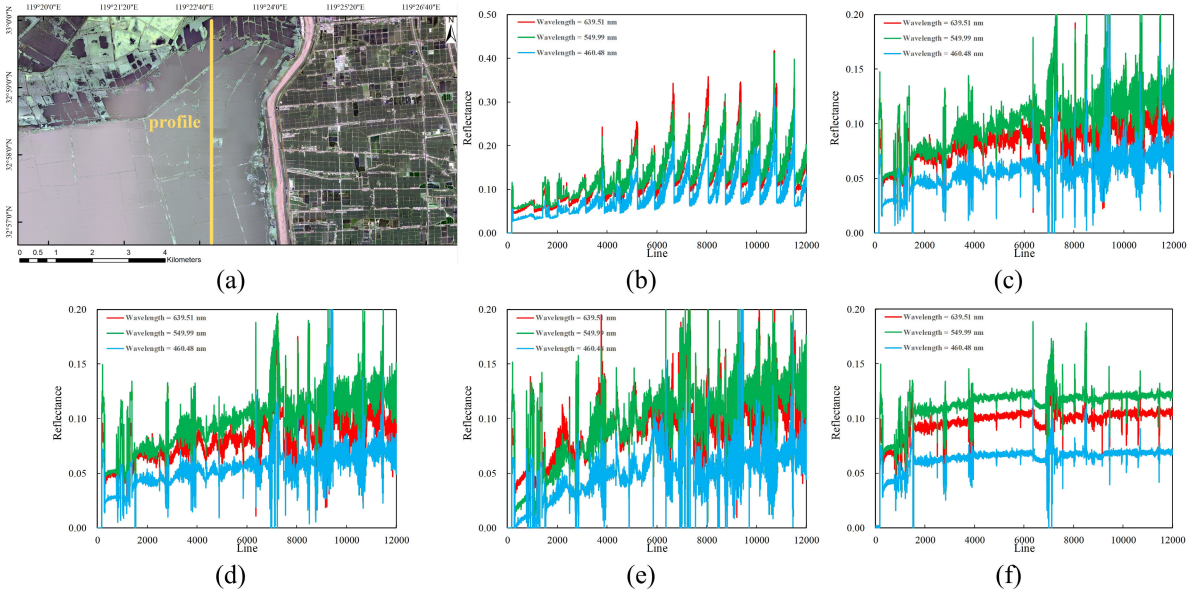


Fig. 8. Reflectance profiles of the red, green, and blue bands across flight strips before and after correction over a water region in the Gaoyou Lake dataset. (a) Location of the profile. (b) Mosaic result for the original image strips. (c) Mosaic result after quartic empirical correction. (d) Mosaic result after traditional kernel-based correction. (e) Mosaic result after semiempirical water BRDF correction. (f) Mosaic result using the proposed method.

the MAD value is significantly lower than the values for the three comparison methods, with the maximum MAD reduced by over 80%.

To further verify the accuracy of the reflectance after BRDF correction, we compared the spectra with the ground-measured results. For water bodies, since the ASD result is remote sensing reflectance (R_{rs}), the image water reflectance was converted into water remote sensing reflectance. Given the inherent fluidity of water bodies, we selected water sampling points for validation. Four water samples were selected for comparison in Gaoyou Lake area. For the corresponding image reflectance spectra, we utilized the coordinates recorded at the ground spectral sampling points as the center pixel. We then selected a 5 $\times$ 5 pixel window centered on each of these points in the image and calculated the mean reflectance for validation.

The water spectral results before and after correction are shown in Fig. 12. The shaded area represents the mean $\pm$ 2 standard deviations (SDs) of the selected image reflectance. Compared with the uncorrected image reflectance data, all four corrected water reflectance results demonstrated a higher degree of consistency with the measured water reflectance. Notably, for the results obtained using the proposed correction method in this study, as shown in Fig. 12(e), it can be seen

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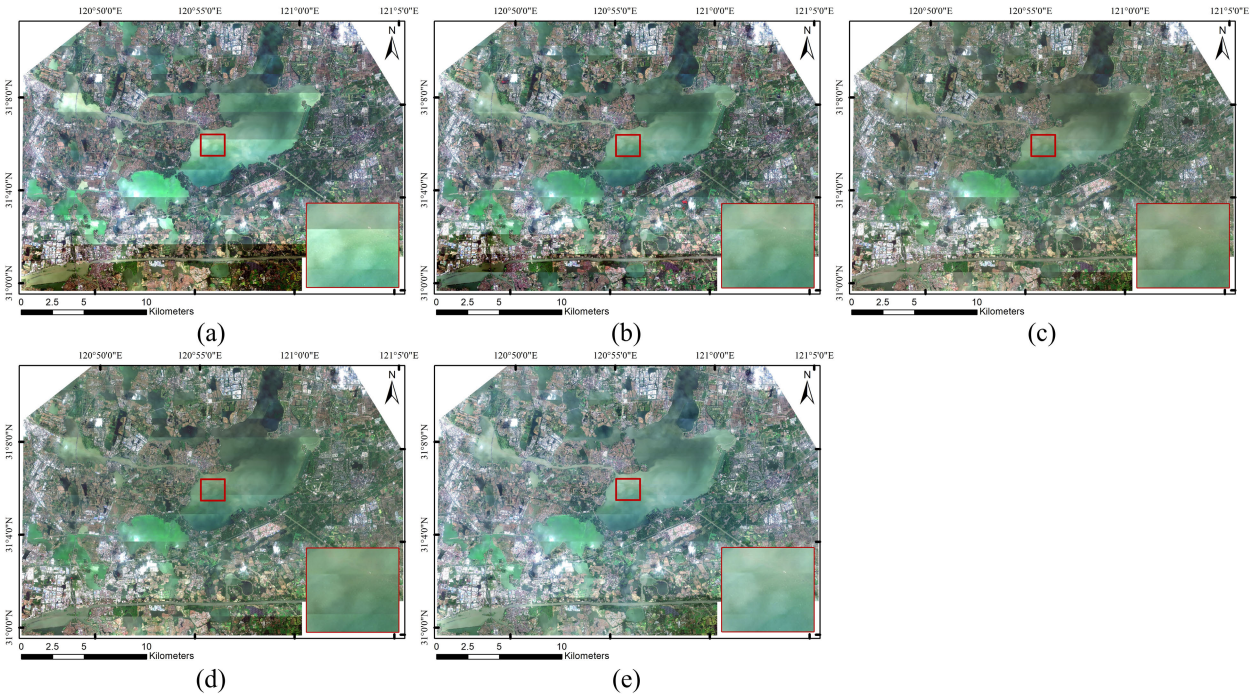


Fig. 9. Image mosaicking results for the 15 image strips of Dianshan Lake. RGB color composition with 639.02, 550.87, and 460.64 nm. (a) Mosaic result for the original image strips. (b) Mosaic result after quartic empirical correction. (c) Mosaic result after traditional kernel-based correction. (d) Mosaic result after semiempirical water BRDF correction. (e) Mosaic result using the proposed method.

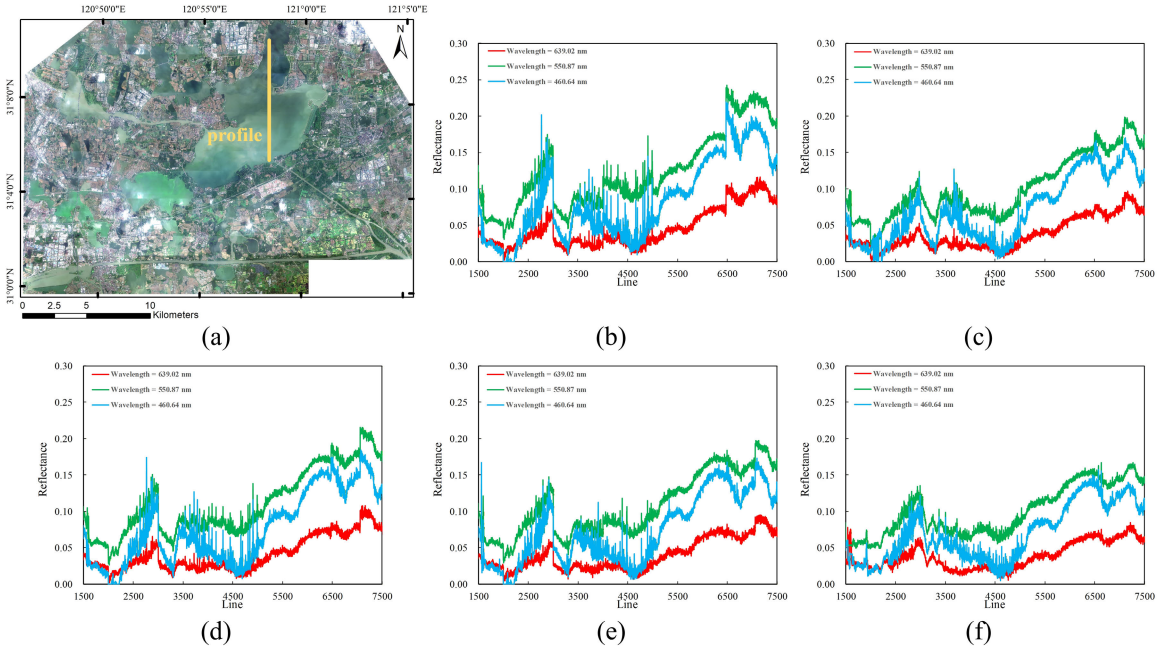


Fig. 10. Reflectance profiles of the red, green, and blue bands across flight strips before and after correction over a water region in the Dianshan Lake dataset. (a) Location of the profile. (b) Mosaic result for the original image strips. (c) Mosaic result after quartic empirical correction. (d) Mosaic result after traditional kernel-based correction. (e) Mosaic result after semiempirical water BRDF correction. (f) Mosaic result using the proposed method.

that the four sets of water spectra have high similarity in the range of 400–1000 nm, with the positions of the absorption and reflection characteristics of the water bodies matching around 550, 670, 700, and 820 nm. Table II presents the accuracy results for water in Gaoyou Lake area. The proposed method in this study demonstrates the best performance in terms of R^2, mse, and SA. The coefficient of determination R^2 is 0.9358, 0.9940, 0.9479, and 0.9399, respectively. The mse is 0.00147, 0.00144, 0.00250, and 0.00101, respectively. The SA is 6.9738, 2.5311, 6.5297, and 8.1925, respectively.

2) Dianshan Dataset: Fig. 11(b) shows the MAD curves for Dianshan dataset. The result is similar to that for Gaoyou dataset, and the corrected image has smaller MAD values. The change in MAD values before and after correction is relatively small due to the absence of a serious sun glint effect. The

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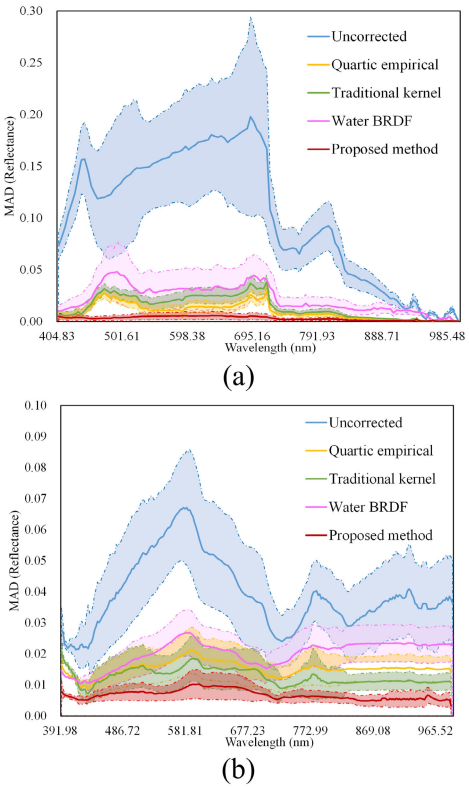


Fig. 11. MAD of the reflectance in each band before and after correction. (a) Gaoyou dataset. (b) Dianshan dataset.

TABLE II
ACCURACY RESULTS IN GAOYOU DATASET

Object	Method	R^2	MSE	SA (°)
Water1	Uncorrected	0.7623	0.00482	11.7813
	Quartic empirical	0.8582	0.00219	7.1240
	Traditional Kernel	0.9326	0.00151	9.4978
	Water BRDF	0.8230	0.00264	7.5282
	Proposed	**0.9358**	**0.00147**	**6.9738**
Water2	Uncorrected	0.9329	0.00482	6.2815
	Quartic empirical	0.9806	0.00259	2.8907
	Traditional Kernel	0.9812	0.00255	3.3565
	Water BRDF	0.9523	0.00406	3.0419
	Proposed	**0.9940**	**0.00144**	**2.5311**
Water3	Uncorrected	0.8538	0.00522	7.2074
	Quartic empirical	0.8685	0.00505	6.7387
	Traditional Kernel	0.8858	0.00417	6.3214
	Water BRDF	0.9110	0.00327	6.8287
	Proposed	**0.9479**	**0.00250**	**6.5297**
Water4	Uncorrected	0.8570	0.00223	12.0928
	Quartic empirical	0.9321	0.00116	8.4742
	Traditional Kernel	0.9201	0.00117	8.9816
	Water BRDF	0.8878	0.00138	9.7820
	Proposed	**0.9399**	**0.00101**	**8.1925**

proposed method achieves an average MAD below 0.01 across all spectral bands.

According to the selection principles of validation points in Gaoyou Lake area, four water samples were selected for comparison in Dianshan dataset, and the results are shown in Fig. 13. The corrected spectra are also highly consistent with the ground-measured spectra, which proves the applicability of the proposed correction method. The accuracy results are listed in Table III. The R^2 values of the four water samples

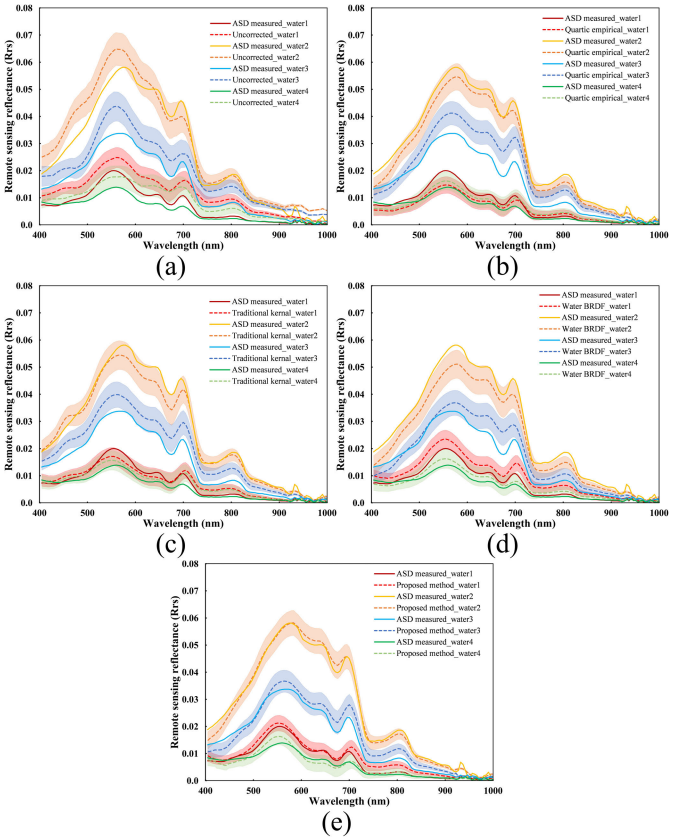


Fig. 12. Spectral contrast diagram for water using different correction methods in Gaoyou Lake area. (a) Atmospheric result without correction. (b) Quartic empirical correction. (c) Traditional kernel-based correction. (d) Semiempirical water BRDF correction. (e) Proposed correction method. The shaded area is two SDs around the mean value.

TABLE III
ACCURACY RESULTS IN DIANSHAN DATASET

Object	Method	R^2	MSE	SA (°)
Water1	Uncorrected	0.7978	0.00432	10.4931
	Quartic empirical	0.9424	0.00189	7.6226
	Traditional Kernel	0.9240	0.00217	7.3167
	Water BRDF	0.8867	0.00264	6.2341
	Proposed	**0.9688**	**0.00130**	**6.0461**
Water2	Uncorrected	0.8500	0.00360	4.7322
	Quartic empirical	0.9816	0.00126	3.9137
	Traditional Kernel	0.9402	0.00227	4.2768
	Water BRDF	0.9756	0.00145	4.0931
	Proposed	**0.9902**	**0.00092**	**3.5634**
Water3	Uncorrected	0.9711	0.00258	3.0996
	Quartic empirical	0.9724	0.00252	5.3614
	Traditional Kernel	0.9564	0.00318	2.3168
	Water BRDF	0.9266	0.00489	2.0243
	Proposed	**0.9960**	**0.00096**	**1.3899**
Water4	Uncorrected	0.9553	0.00205	5.4364
	Quartic empirical	0.9831	0.00126	4.1370
	Traditional Kernel	0.9835	0.00124	3.6594
	Water BRDF	0.9787	0.00141	4.0453
	Proposed	**0.9869**	**0.00110**	**3.6181**

are 0.9688, 0.9902, 0.9960, and 0.9869, respectively. The mse is 0.00130, 0.00092, 0.00096, and 0.00110, respectively. The SA is 6.0461, 3.5634, 1.3899, and 3.6181, respectively.

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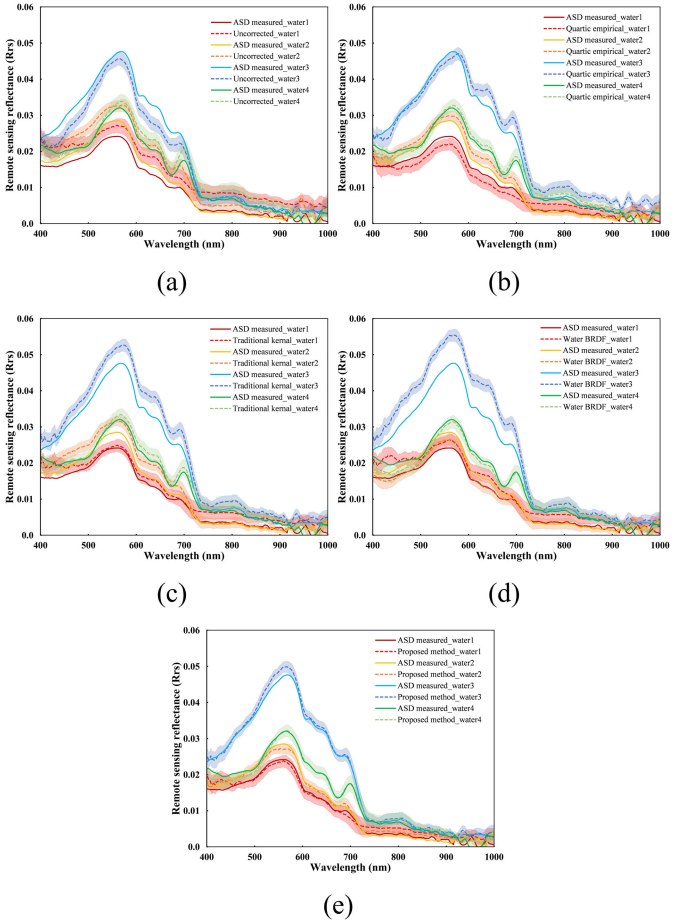


Fig. 13. Spectral contrast diagram for water using different correction methods in Dianshan Lake area. (a) Atmospheric result without correction. (b) Quartic empirical correction. (c) Traditional kernel-based correction. (d) Semiempirical water BRDF correction. (e) Proposed correction method. The shaded area is two SDs around the mean value.

3) BRDF Characteristics in Water-Land Scenes: Taking the water-land mixed scenes considered in this study as an example, when flying over the lake surface and scanning for imaging, the radiation difference caused by sensor attitude between adjacent scan lines can reach 20%–30%, which is manifested in the original mosaicked image shown in Fig. 7(a) as an obvious light and dark block phenomenon. However, for land objects, there are no obvious brightness differences. The main reason for this phenomenon is the influence of water body specular reflection, which causes changes in the radiation energy entering the sensor through specular reflection due to changes in the attitude angle. In addition, the radiation signal of water bodies is much weaker than that of land objects, and the change in attitude intensifies both the scattering effect and the adjacency effect [34]. In this study, we found that the different flight directions between strips also bring reflectance differences, which are particularly obvious on water bodies. Due to the anisotropy of the radiance field just above the water surface outside the sun glint shown by water bodies, the mirror reflection of the water body and the possible hotspot effect exacerbate the anisotropy between strips, making the signal intensity received by the sensor significantly different.

On top of the traditional semiempirical kernels, this study adds a Cox–Munk kernel K_{spec} to model the water-surface specular term. To further verify the contribution of this kernel, we fit the reflectance data in glint-affected water areas using the standard Ross–Li semiempirical model without K_{spec} and the proposed model and compared their BRDF fitting performance. As shown in Fig. 14, we plotted the fit BRDF curves of reflectance for the red, green, and blue bands as a function of view zenith angle. The standard model cannot accurately represent how sun glint varies with the view zenith angle and fails to capture the sun glint peak, leading to an underestimation in the specular direction and a pronounced overestimation in the weak-glint region at view angles greater than 10°. In contrast, the proposed model with K_{spec} accurately fits the water reflectance under sun glint conditions. Quantitatively, for the standard model, the fitting accuracy R^2 for the red, green, and blue bands is 0.9183, 0.9075, and 0.9072, respectively, and the corresponding mse values are 0.0795, 0.0812, and 0.0761. For the proposed model with K_{spec}, the R^2 values for the red, green, and blue bands are 0.9437, 0.9328, and 0.9587, respectively, and the corresponding mse values are 0.0762, 0.0771, and 0.0701. The fitting accuracy coefficient R^2 increases by about 4%, while the mse is reduced by approximately 5% in the glint-affected angular range. The quantitative comparison of BRDF fitting performance is summarized in Table IV. These results provide evidence that K_{spec} improves the accuracy of BRDF function modeling and effectively mitigates residual sun glint effects in water-dominated scenes.

TABLE IV
QUANTITATIVE COMPARISON OF BRDF FITTING PERFORMANCE FOR SUN GLINT-AFFECTED WATER REFLECTANCE ACROSS RGB BANDS

Band	Model	R^2	MSE
Red	Traditional Kernel without K_{spec}	0.9183	0.0795
	Proposed Model with K_{spec}	**0.9437**	**0.0762**
Green	Traditional Kernel without K_{spec}	0.9075	0.0812
	Proposed Model with K_{spec}	**0.9328**	**0.0771**
Blue	Traditional Kernel without K_{spec}	0.9072	0.0761
	Proposed Model with K_{spec}	**0.9587**	**0.0701**

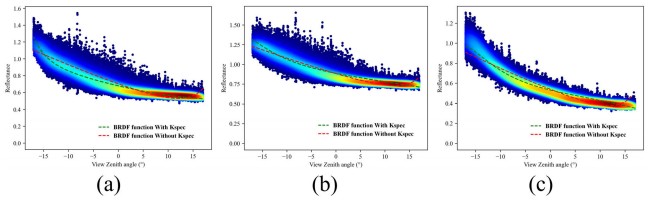


Fig. 14. Comparison of BRDF fitting performance for sun glint-affected water reflectance in (a) red, (b) green, and (c) blue bands as a function of view zenith angle.

4) Practical Implications and Future Work: The sun glint in water bodies directly affects the results of the subsequent BRDF correction [35]. Therefore, for a mixed water and land scene, the BRDF characteristics of water bodies and land surfaces should be considered separately and corrected accordingly. For water areas, which can be significantly affected by sun glint, their image reflectance can reach several times or even tens of times the actual water reflectance, seriously affecting the uniformity of the water body. When performing BRDF correction on a water body, the sun glint should be precorrected, or the impact of sun glint should be considered in the correction, in order to reduce the error caused by

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non-water-body radiation signals on BRDF correction [5]. Such correction is also important for quantitative water-quality inversions, because parameters such as chlorophyll-a, suspended matter, and turbidity are highly sensitive to BRDF effects and stripwise reflectance biases [36].

From an operational perspective, proper flight design is crucial. Theoretically, when the solar elevation angle is greater than 90°-FOV/2, the light from the sun will reflect onto the CCD, causing sun glint in the image. Therefore, the solar elevation angle needs to be considered during flight planning to avoid large-area sun glint. Furthermore, during the flight, other factors such as long acquisition time spans and significant topographic relief can also influence BRDF correction [16]. In this study, these factors were not considered in the model as the terrain was predominantly flat and the time interval between adjacent flight strips was relatively short. However, this simplification may not be applicable to mountainous reservoirs, narrow rivers, or other water bodies surrounded by complex terrain. In addition, some vegetation-focused studies employ fine-grained species classification to refine BRDF correction [15], [23]. However, the main object of this study is inland water bodies, and the optical properties of these water bodies are relatively consistent. Therefore, no classification of water body optical properties was conducted. Another limitation is that the proposed attitude normalization approach relies on accurate and temporally synchronized sensor attitude data. Uncertainties in roll and pitch measurements may affect the estimation of the sensor zenith angle and the line-by-line normalization coefficient, while heading errors may influence the relative azimuth angle and the subsequent kernel-based BRDF correction. Therefore, inaccurate synchronized attitude records may lead to residual stripwise biases or incomplete correction of attitude-induced brightness variations.

In future work, large-scale or long-duration acquisitions over complex terrain should incorporate illumination and terrain normalization strategies—for example, by introducing illumination correction factors and DEM-based topographic correction methods—as well as exploring more refined classification schemes to further improve the robustness and accuracy of BRDF correction under diverse and complex conditions. In addition, uncertainty propagation and sensitivity analysis of attitude parameters should be further considered to improve the reliability of the correction framework.

V. Conclusion

In this article, we have proposed an integrated BRDF correction framework for airborne hyperspectral data with normalized multiple imaging conditions for inland water bodies, achieving seamless mosaicking for multiple image strips and obtaining accurate reflectance data. We focused on the impact of sensor attitude changes and flight direction on the BRDF effects, developed a sensor attitude normalization correction method, and proposed a kernel-based semiempirical inland water correction model to eliminate the brightness differences caused by different imaging conditions, effectively reducing the uncertainty caused by BRDF effects. Experiments across multiple sensor platforms further validate the robustness and applicability of the proposed method.

Currently, with the development of spectral imaging technology, imaging spectrometers have become more compact and lighter in weight, which has promoted the progress of UAV-borne hyperspectral acquisition systems. However, UAVs are lighter in weight and are faced with more serious attitude changes. Therefore, the calibration framework considering multiple imaging conditions proposed in this article has wide application prospects for the current development trend of lightweight and miniature airborne imaging systems and can provide more reliable reflectance data for subsequent quantitative water-quality retrievals.

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